FINAL PROJECT REPORT

3D Semantic Vector Space Explorer

Spatial Visualization of Knowledge Bases for RAG Systems

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Abstract

Retrieval-Augmented Generation (RAG) systems decide which documents an LLM sees by comparing high-dimensional text embeddings, a step that is invisible to users. This project builds an interactive 3D visualization that makes that step visible. A reproducible Python pipeline embeds 270 documents from the 20 Newsgroups corpus with all-MiniLM-L6-v2, projects them to 3D with UMAP, clusters them with K-Means and derives semantic (cosine) and structural (source category) relations. A static WebGL application renders the result as a force-directed graph with a perceptually motivated encoding: Tableau10 color for topic clusters, square-root-scaled sphere size for document length and a dual opacity-plus-width encoding for similarity. Clicking a document simulates a RAG query: the eight nearest neighbors light up as a gold "query star", the rest of the corpus fades into context, and a side panel shows the retrieved texts with their cosine scores.

The application runs fully offline at 125+ FPS on a mid-range laptop, passes an identical 10-step scenario in Chrome, Firefox and Safari, and was evaluated with seven outside participants. Beyond the artifact itself, the visualization exposed two properties of embedding retrieval that are easy to state and hard to see: semantic neighbors are not spatial neighbors, and short, generic queries retrieve scattered, cross-topic context.

1 Introduction and Motivation

Modern LLM applications increasingly rely on RAG [1]: a user query is embedded into a vector space, the nearest documents of a knowledge base are retrieved, and these documents are passed to the model as context. The quality of the final answer depends heavily on this retrieval step, yet the criteria for "semantically similar" are hidden in vectors with hundreds of dimensions. For most users, and often for developers, retrieval is a black box.

The goal of this project is an Explainable-AI (XAI) visualization tool that opens that box. Instead of explaining retrieval with formulas, the tool lets the user *see* it: the knowledge base becomes an explorable 3D space in which proximity reflects meaning, clusters reflect topics, and a single click reproduces what a retriever does for a query.

The project follows Shneiderman's Visual Information-Seeking Mantra, "overview first, zoom and filter, then details on demand" [6]. The overview is the full 3D map of the corpus; zoom and filter are provided by camera controls, cluster toggles, an edge-type switch and a similarity threshold; details on demand come from tooltips and a document side panel. The central interaction, Focus+Context [7], is reinterpreted as a retrieval simulation: the selected document is the query, its top-k neighbors are the retrieved context, and everything else becomes dimmed background.

The final deliverables are a deterministic data pipeline (Python), a static web application (HTML/JS/CSS, no build step, fully offline), this report and a short demo. The rest of the report describes the data pipeline (§2), the visual encoding and its perceptual justification (§3), the interactions (§4), the evaluation (§5) and the limitations (§6).

2 Data and Pipeline

The backend is a single script, `build_graph.py`, driven by one configuration file (`config.py`). It runs six stages: corpus loading, embedding, 3D projection, clustering, edge construction and JSON export. Each stage caches its result keyed by a hash of the corpus, so later stages can be tuned without recomputing earlier ones. The output, `graph_data.json`, is the contract between backend and frontend; it was frozen before any frontend work started and changed only once, as an explicitly versioned event (schema v1.1, §4.2).

2.1 Corpus

The corpus is the 20 Newsgroups dataset [10], chosen for its clear topical structure and zero acquisition effort. Six topically distant categories were selected: *sci.space*, *rec.sport.hockey*, *comp.graphics*, *sci.med*, *rec.autos* and *talk.politics.mideast*. Headers, footers and quoted replies were removed (otherwise clustering "discovers" e-mail headers instead of semantics), documents were filtered to 200–3000 characters after cleaning, and duplicates were removed on normalized text. A seeded sample of 45 documents per category gives 270 documents, perfectly balanced. Mean document length ranges from 595 characters (comp.graphics) to 999 characters (talk.politics.mideast).

2.2 Embeddings

Each document is embedded with the sentence-transformers model all-MiniLM-L6-v2 [2] into a 384-dimensional space, with `normalize_embeddings=True`. For unit vectors cosine similarity equals the dot product, so the full similarity matrix is a single matrix product. The model truncates input at 256 word pieces;

for a corpus limited to 3000 characters this was accepted as a deliberate trade-off: for the longest posts the embedding represents the beginning of the text, while sphere size still reflects the full length. A smoke test confirmed semantic sanity before use (two related sentences: cosine 0.531; unrelated pair: −0.005).

2.3 Projection to 3D

Coordinates come from UMAP [3] (`n_components=3`, `n_neighbors=15`, `min_dist=0.1`, cosine metric, fixed seed). UMAP was chosen over PCA because it preserves local neighborhoods, which is exactly the property a retrieval visualization must show, and over t-SNE because it also keeps more global structure and is deterministic with a seed. A 2D control projection colored by the **true** category labels, which the model never sees, verifies that the embedding and the projection preserve topical structure (Figure 1). Five of the six categories form compact, well-separated groups; hockey and Middle-East politics are isolated almost perfectly. sci.space is the most diffuse category and overlaps with sci.med and rec.autos, which anticipates the clustering result in Table 1. Because the acceptance criterion set in advance (at least five of six categories clearly separated) was met, the default parameters were kept and no parameter search was needed. Coordinates are centered and rescaled to a standard deviation of about 40 scene units, a size that fits the default camera.

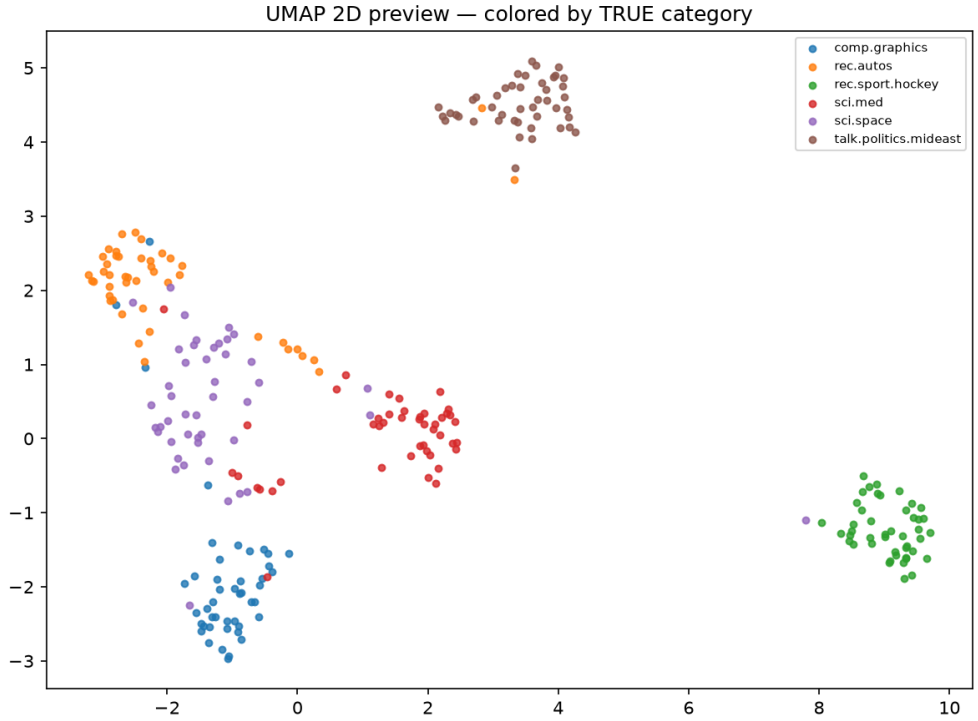


Figure 1. UMAP 2D control projection of the 270 documents, colored by the true 20 Newsgroups category (not used by the model). Five of six categories form separated groups; sci.space overlaps with sci.med and rec.autos.

2.4 Clustering and cluster labels

K-Means is run on the original 384-dimensional vectors, not on the UMAP coordinates, because the projection distorts distances. The number of clusters was selected by the cosine silhouette score over k = 4…10 (Figure 2). The maximum is at k = 6 (silhouette 0.144), matching the six source categories. Absolute silhouette values are modest, as is typical for short, noisy forum posts, but well above the pre-defined fallback trigger (0.05) that would have switched the corpus to arXiv abstracts.

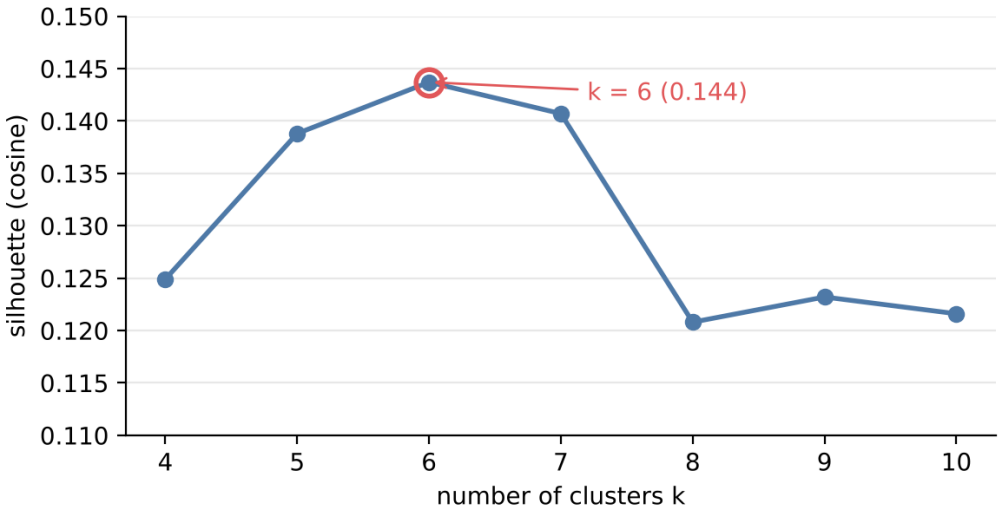


Figure 2. Silhouette score (cosine) of K-Means on the 384-D embeddings for k = 4…10. The maximum at k = 6 matches the six source categories.

Table 1 compares clusters with the true categories. The matrix is strongly diagonal: hockey is recovered perfectly (45/45), and the other categories reach 36–43 documents in their dominant cluster. Most "errors" are semantically plausible, for example medical or automotive posts that discuss space-related or technical topics. For the legend, each cluster is labeled with its top TF-IDF terms instead of a number ("game, team, hockey, nhl" is more informative than "Cluster 4").

Table 1. Contingency matrix: K-Means clusters (rows) vs. true 20 Newsgroups categories (columns).

Cluster (TF-IDF label)	graphics	autos	hockey	med	space	mideast
0 — space, earth, like, long	4	5	0	6	**37**	2
1 — israel, jews, people, israeli	0	1	0	0	0	**43**
2 — pain, don, doctor, course	1	1	0	**36**	3	0
3 — does, thanks, card, image	**38**	0	0	2	1	0
4 — game, team, hockey, nhl	0	0	**45**	0	0	0
5 — car, dealer, cars, price	2	**38**	0	1	4	0

2.5 Semantic edges and threshold calibration

Semantic edges connect documents with high cosine similarity. The proposal assumed a priori thresholds of 0.6 (default) with width tiers at 0.6/0.7/0.85. Inspecting the empirical distribution showed that these values do not fit the data: over all document pairs the median cosine is only 0.025, the 90th percentile 0.178 and the 99th percentile 0.411. At 0.6 only 16 edges would remain, leaving an almost empty graph and a dead slider. The distribution (Figure 3) shows why. The thresholds were therefore recalibrated to the data: edges are exported from cosine ≥ 0.25 and the default UI threshold is 0.35. This is the most important deviation from the proposal, and it generalizes: similarity thresholds are properties of a model–corpus pair, not universal constants.

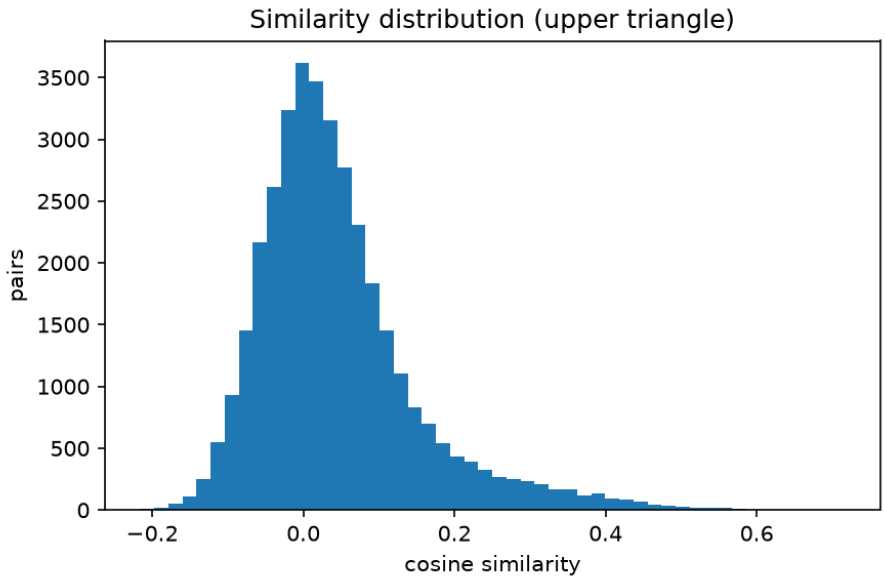


Figure 3. Distribution of cosine similarity over all 36,315 document pairs. The mass lies around zero (median 0.025); the proposal's threshold of 0.6 falls in the extreme tail, which motivated the recalibration to 0.25 / 0.35.

To avoid a "hairball", an exported edge must also be among the top-10 strongest edges of at least one endpoint. Figure 4 shows the resulting edge budget: 661 semantic edges at the default threshold, inside the target window of 300–1500 that keeps the cluster structure readable.

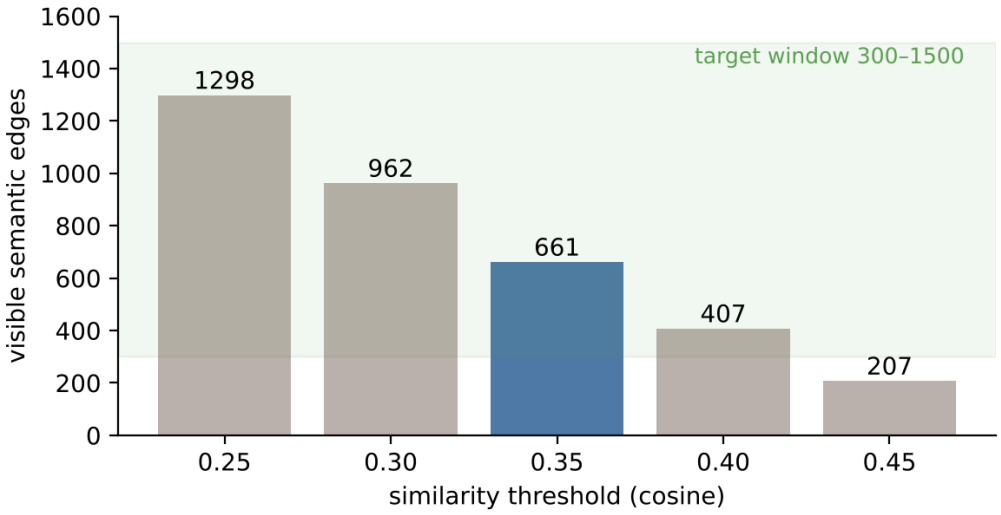


Figure 4. Number of visible semantic edges per similarity threshold (top-10 guard applied). The default 0.35 lies in the 300–1500 target window.

Every node also stores its top-10 neighbor list with cosine values. This precomputed list turns the retrieval simulation in the frontend into a constant-time lookup, with no client-side similarity computation.

2.6 Structural edges

Structural edges encode metadata: "same source category". Connecting all documents of a category would create cliques of about 1000 edges each, so each node is linked only to its three most similar documents of the same category, forming a category skeleton. After the threshold recalibration most of these pairs already existed as semantic edges. Instead of dropping them, such edges are dual-typed (a semantic edge carrying a `struct` flag). The category skeleton contains 601 edges (2.23 per node), of which 15 are structural-only in the final schema.

2.7 JSON contract, schema v1.1 and determinism

The JSON file contains a `meta` block (corpus, k, threshold range, cluster labels), the nodes (id, title, category, cluster, token count, coordinates, trimmed text, neighbor list) and the links (source, target, value, type, optional flags). The frontend builds its legend and slider range from `meta`, never from hardcoded values. A separate validator checks referential integrity, value ranges, required fields, cluster labels and, since schema v1.1, the types of the optional flags.

Schema v1.1 added a guarantee that every node-to-neighbor pair exists as a link (flag `knn`), the result of a bug found during interaction design (§4.2). The final file has 270 nodes, 1868 semantic and 15 structural-only links, and a size of 0.44 MB. Determinism was verified twice: repeated full rebuilds from scratch (`--recompute`) produce byte-identical files (identical SHA-256) for both schema versions. A full rebuild takes about 80 seconds.

3 Visual Encoding

Every visual channel was assigned according to the type of data it carries and the perceptual properties of the channel selection [9] (Table 2). Figure 5 shows the overview with all encodings active.

Table 2. Mapping of data attributes to visual channels.

Data attribute	Type	Visual channel	Justification
Topic cluster	Categorical (6)	Hue — Tableau10 palette	Hue is pre-attentive for categorical discrimination [4]; Tableau10 is designed for distinguishable categories
Document length	Quantitative	Sphere size, sqrt scale (radius 2–8)	Perceived size follows area, not radius [5]; sqrt keeps area proportional to value
Cosine similarity	Quantitative	Edge opacity (continuous) + width (3 tiers)	Redundant dual encoding: luminance contrast for fine order, width for coarse classes
Relation type	Categorical (2)	Edge hue: blue (semantic) vs. gray (structural)	Separates relation types without competing with cluster hues
Semantic position	Spatial	3D position (UMAP seed + gentle forces)	Position is the most accurate channel [5]; proximity = similarity
Retrieval result	Selection	Gold edges, dimming, enlarged query node	Focus+Context: strong contrast for the subgraph, context preserved [7]



Figure 5. Overview of the corpus at the default threshold 0.35 (676 visible edges): six color-coded topic clusters, sphere size by document length, blue semantic edges with opacity and width by similarity, and the filter panel with TF-IDF cluster labels.

3.1 Nodes: color and size

Nodes are rendered as lit 3D spheres (`MeshLambertMaterial`) under the default ambient and directional lights. Directional shading gives each sphere a visible gradient, which provides depth cues in the 3D scene without custom shaders. All 270 spheres share one unit geometry scaled per node, which keeps GPU memory low.

Color: Cluster membership is categorical, and hue is the strongest pre-attentive channel for categorical discrimination [4]. The Tableau10 palette [8] was chosen because its colors are designed to be distinguishable from each other on screen. The same color scale feeds the spheres, the legend and the side panel, so the mapping cannot drift between views.

Size: Document length is quantitative. Following Cleveland and McGill [5], viewers judge the area of a circle rather than its radius, so a linear radius mapping would exaggerate long documents: four times the length would look sixteen times larger. A square-root scale (`d3.scaleSqrt`, radius 2–8) keeps perceived area proportional to length. Size is the least accurate of the channels used, which is acceptable because length is secondary information; the user study (§5.3) confirmed that it is also the most ambiguous one, which led to an explicit size legend.

3.2 Edges: multi-relational, dual-encoded

Semantic similarity is encoded twice. Opacity varies continuously, $\alpha = 0.15 + 0.65 \cdot (\text{value} - 0.25) / 0.45$, so stronger relations appear brighter against the dark background; width varies in three tiers (thresholds 0.35/0.45/0.55, widths 0.5/1.2/2.5), recalibrated from the proposal's 0.6/0.7/0.85 to the empirical similarity range. The redundancy makes the strongest relations stand out pre-attentively while weak ones remain visible as faint context. Structural edges use a neutral gray at a fixed thin width, so the relation type is separable from similarity strength (Figure 6).

A technical finding affected this encoding. Per-edge transparency was implemented through the alpha channel of RGBA colors (verified with a high-contrast test color). The edges nevertheless looked washed out

until it turned out that the library applies a global link opacity of 0.2 that multiplies every per-edge alpha. Setting it to 1 restored the designed range and made the gold focus highlight clean.

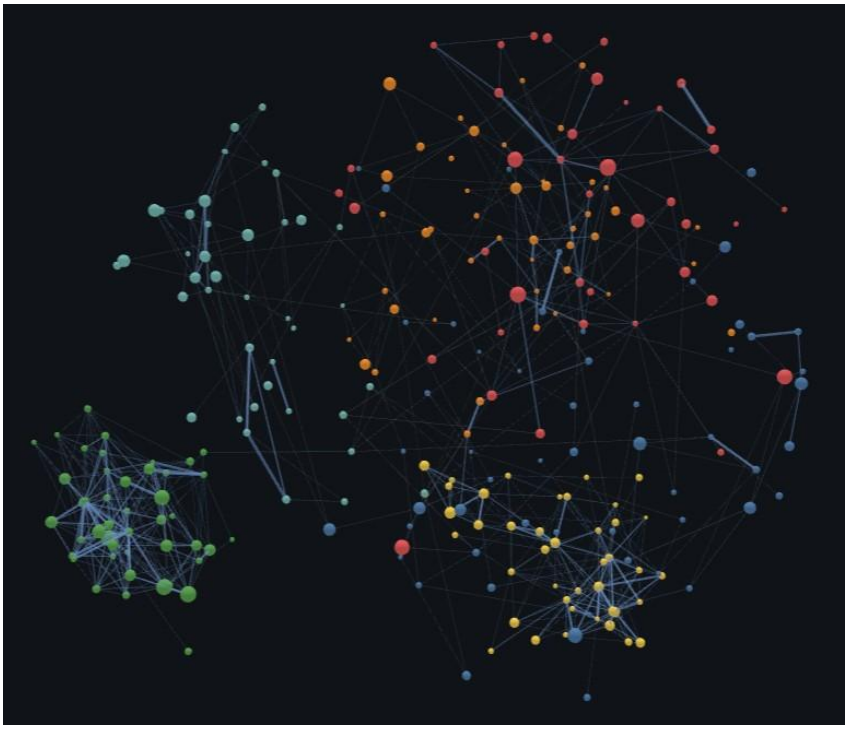


Figure 6. Edge encoding: thick, bright edges inside dense clusters (high similarity) versus faint thin edges between clusters (low similarity).

3.3 Layout

The layout combines a precomputed projection with a light physical simulation. Node positions are seeded from UMAP, so clusters are separated from the first frame. The initially planned forces (link strength 0.1 · value, charge −30) visibly collapsed this separation, so they were recalibrated to gentle physics: link strength 0.03 · (value − 0.25) / 0.45 rescaled to the empirical range, charge −10, and a 15-second cooldown after which the simulation freezes. The layout settles within seconds while preserving the UMAP structure. The visible overlap between the science and computer-graphics clusters reflects real semantic proximity, not a layout artifact.

3.4 Perceptual quality control

Hairball test: At the default threshold the cluster structure is readable from the default camera position, so no further edge reduction was needed.

Color-vision deficiency: Simulating deuteranopia and protanopia (Chrome DevTools) showed that yellow, teal and blue stay separable, but red, orange and green collapse into a similar brown-olive range. No categorical palette with six hues is fully robust to these conditions. Tableau10 was kept and the design relies on redundant channels [4]: clusters are also separated spatially, and the legend names each cluster with keywords. This is documented as a limitation (§6).

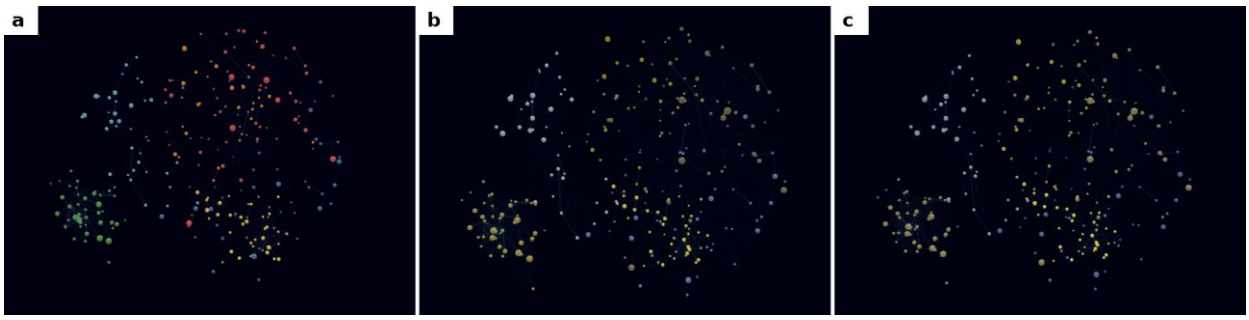


Figure 7. Color-vision deficiency check: (a) normal vision, (b) simulated deuteranopia, (c) simulated protanopia. Red, orange and green clusters converge to similar olive tones; spatial separation remains.

Background. Pure black (#000000) and dark graphite (#101418) were compared. On the laptop display the difference was imperceptible; graphite was chosen because it matches the loading screen and is less prone to uneven black levels on projectors.

3.5 Design evolution

The final encoding is the result of several iterations, each driven by a concrete observation rather than taste (Figure 8). The first prototype only verified that WebGL rendering worked (a). The first render of the real data used the library defaults: automatically assigned colors and uniform gray edges (b). Introducing Tableau10 spheres made the clusters identifiable, but with the default force strengths and all edges drawn alike the scene degenerated into a hairball in which the UMAP separation collapsed (c). Recalibrating the forces to the empirical similarity range, encoding similarity in edge opacity and width, and removing the library's global edge opacity produced the final scene (d), in which both the cluster structure and the strongest relations are readable at a glance.

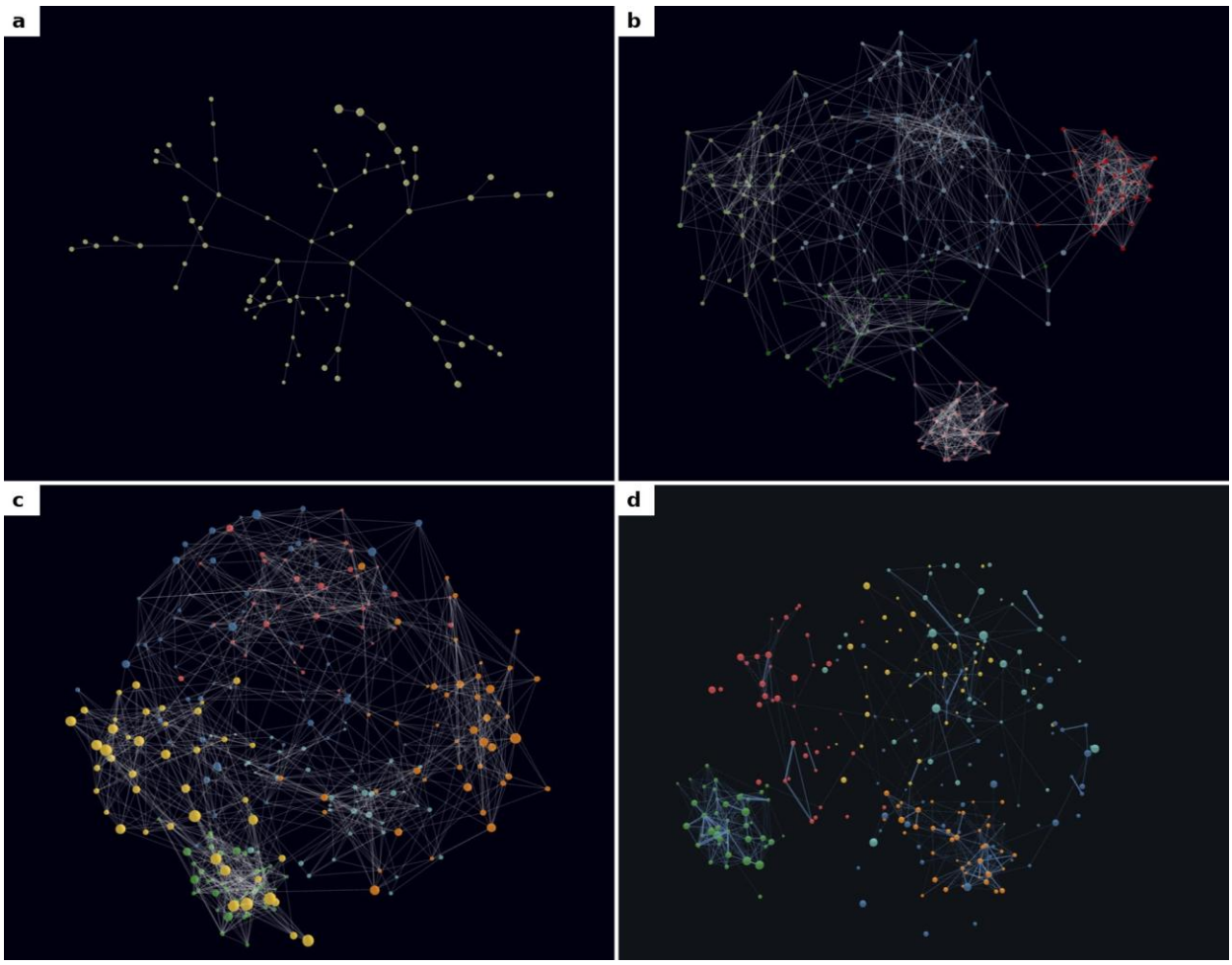


Figure 8. Evolution of the scene: (a) WebGL smoke test with a sample tree; (b) first render of the real data with library defaults; (c) Tableau10 spheres with uniform edges and default forces, a hairball; (d) final encoding.

4 Interactions

4.1 Architecture: one state, one render rule

All interactions share one state object (mode, focused node, its neighbors, hidden clusters, edge view, similarity threshold) and one function, `applyState()`, which recomputes the visibility and style of the whole scene after every change. No interaction modifies the scene directly. This discipline removes a whole class of combination bugs by construction, such as a graph that stays dimmed after an exit. Filters change only visibility and never rebuild the graph data, so the layout never restarts while the user filters. The interplay of features follows explicit rules:

1. Filters apply first: a query highlights only within the filtered subgraph.
2. Changing filters during a query keeps the query active and recomputes visibility.
3. If the query document itself is filtered out, the query is cleared and the view returns to the overview.
4. Hiding a cluster overrides everything; query dimming overrides normal styling.

4.2 Focus+Context: simulating a RAG query

Clicking a document starts a simulated retrieval. The document becomes the query; its eight nearest neighbors by cosine similarity (read directly from the precomputed list) become the retrieved context. All other documents and edges are dimmed to an opacity of 0.08 and 0.03, the query-to-context edges are drawn as thick gold lines, the query node is enlarged so it stays visible at the hub, the camera flies to the

subgraph in one second, and a label states the meaning of the scene: *Query: "…" → retrieved 8 context documents*. Esc, a background click, the panel's close button or the Reset button return to the overview (Figure 9).

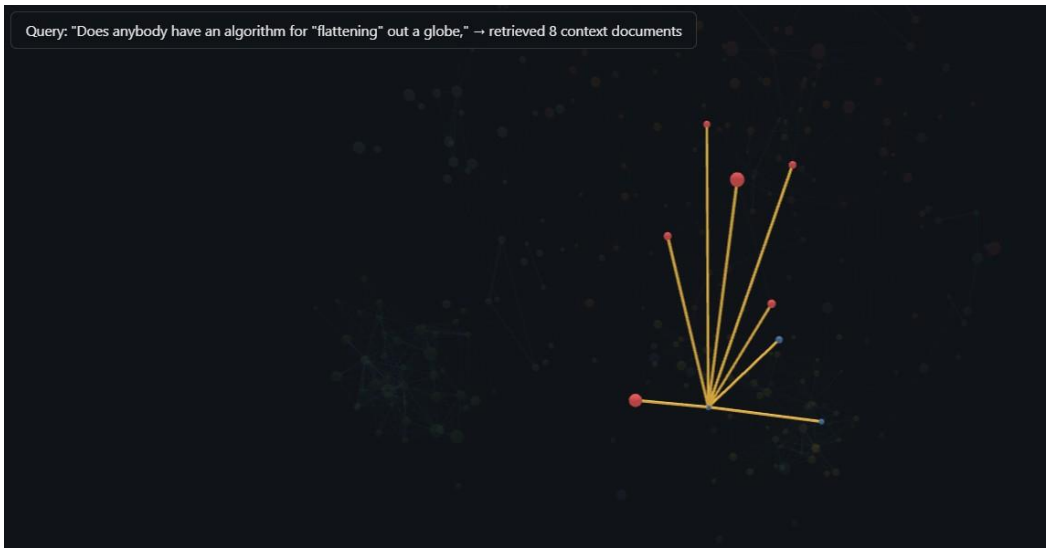


Figure 9. RAG query simulation for document 112, a comp.graphics question about "flattening out a globe". The eight retrieved documents (gold edges) come from two different clusters: retrieval follows meaning, not category.

Two non-obvious problems shaped this interaction and are instructive about embedding retrieval itself.

A retrieval result is not a graph subset: Initially some query stars showed only one or two of the eight edges. A diagnosis for document 112 showed that four of its eight neighbor pairs had no edge in the data at all (cut by the export threshold and by the top-k guard, which is asymmetric: B can be in A's top-8 while A is not in B's top-10) and three more fell below the UI threshold. The fix was a versioned schema change (v1.1): the pipeline guarantees that every neighbor pair exists as a link (flag `knn`, +573 links), and in query mode the star bypasses the threshold and edge-type filters. These links are display-only and exert no force, so the layout is unchanged.

Semantic neighbors are not spatial neighbors: Fixed camera distances failed: depending on the query, the star was either cropped or tiny. Measuring the distances showed why: for one query the eight nearest neighbors in cosine space lay between 52 and 142 scene units from it, a spread of almost 3×. UMAP preserves neighborhoods only approximately, so the camera now measures the actual radius of the retrieved subgraph and backs off by 3× that radius along the current viewing direction, a factor derived from the camera's 40° vertical field of view ($r / \tan 20° \approx 2.75\ r$, plus a margin). The visualization thus makes a property of dimensionality reduction directly visible.

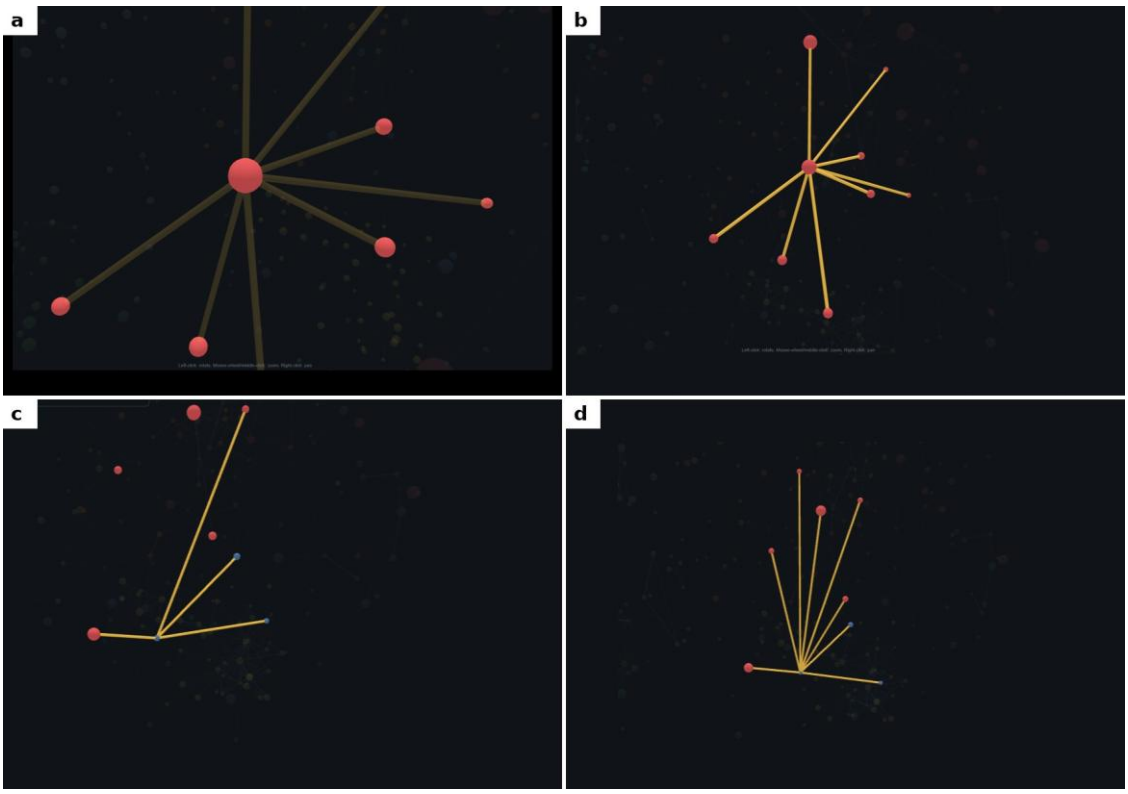


Figure 10. Iterations of the query view. Top: before (a) and after (b) removing the library's global edge opacity and introducing the adaptive camera; the gold edges become clean and the whole star fits the frame. Bottom: the query star of document 112 before (c) and after (d) schema v1.1; four of the eight retrieved neighbors had no edge in the data.

4.3 Layer filtering

A floating panel provides three filters: color-coded cluster checkboxes generated from the data (with TF-IDF labels and document counts), an edge-type switch (both / semantic / structural) and a similarity slider with a live count of visible edges. The slider range comes from the JSON contract; its maximum was set to 0.75 because the data contains almost no edges above 0.7. In the structural view the slider is disabled with a short explanation, because "same category" is metadata, not similarity, and the full 601-edge skeleton is always shown. An empty state with a "Reset filters" button replaces a black screen when all clusters are hidden.

4.4 Details on demand

Hovering a node shows a tooltip with title, category, cluster label and length. Clicking opens a side panel with the full document text and the retrieved-context list with cosine scores, colored like the gold edges (Figure 11). Clicking an entry runs the same query procedure for that document, which enables **semantic navigation** from document to document. Entries whose cluster is hidden are grayed out and not clickable. All document text is inserted as plain text, because newsgroup posts contain characters such as "<" and ">".



Figure 11. Details on demand: side panel with the query document and its eight retrieved documents ranked by cosine similarity; each entry can be clicked to continue the exploration.

4.5 Onboarding

An "About" overlay opens on the first visit and later via a button. It explains in a few lines what the spheres, colors, sizes and lines mean, frames the click as a RAG query and lists the controls (Figure 12). Its content was driven by the user study: because size was the least understood channel, it states explicitly that *size = document length, not importance*, and a size legend with the shortest, median and longest document was added to the panel.

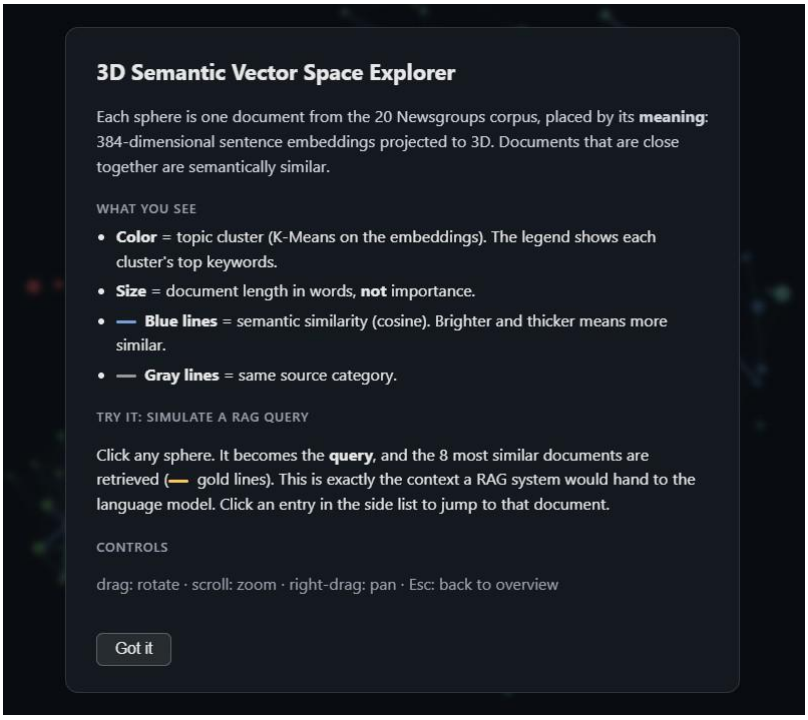


Figure 12. About overlay shown on the first visit: encoding, the RAG framing of a click, and controls.

5 Evaluation and Discussion

5.1 Performance

Frame rates were measured with the Chrome FPS meter on a laptop with a GTX 1650 Ti (Table 3). The first measurement gave only 33–36 FPS while orbiting, low enough to trigger the pre-defined fallback of reducing the number of edges. The cause was not the data: the browser was running on the integrated GPU of the hybrid-graphics laptop. After assigning the browser to the dedicated GPU the same scene ran three times faster, so no data was cut. This is now an item on the demo checklist.

Table 3. Measured frame rates (FPS, Chrome). Target: ≥ 50 on the development machine, ≥ 30 on weaker hardware.

Configuration	At rest	Orbit	Close zoom
Plain lines, integrated GPU	95–100	33–36	130
Plain lines, dedicated GPU	140–144	100–105	144
+ lit spheres	~140	~90	—
+ edge cylinders (width tiers)	—	~95	—
Final (+573 knn links, panels)	—	125+	—

5.2 Robustness and compatibility

Fourteen combinations of interactions were tested systematically (for example: query plus hiding the query's own cluster; query plus slider; rapid query switching; navigation followed by filtering; hiding and re-showing all clusters). All 14 passed after two fixes: re-shown nodes appeared at full brightness during a query, because the library recreates their meshes after the state is applied (fixed by a single opacity rule used both when styling and when creating meshes), and the side panel did not close on one exit path. A 10-minute free-exploration session produced no console errors. An identical 10-step scenario passed in Chrome, Firefox and Safari. The application uses no CDN at runtime (all libraries are vendored) and was verified to load with the network disabled. A fresh clone set up only from the README reproduced the data and the application without any fix.

5.3 User studies

Three small studies with seven different participants, a mix of people with an AI background and people without a technical background, evaluated perception, usability and onboarding (Table 4).

Table 4. User studies.

Study	Participants	Procedure	Result
Perceptual (cold)	5	Graph shown without any explanation; asked what colors, sizes and lines mean	Lines read as relations: 5/5. Colors read as groups/topics: 4/5. Sizes read as document length: 3/5
Interaction	1 (non-technical)	One-sentence introduction, 3 minutes of free exploration	Found all features unaided: query, reading documents, filters, neighbor navigation, return to overview
Onboarding	1 (new to the project)	About overlay only, no help	Discovered the query interaction in under one minute; described the tool as understandable

The perceptual results follow the ranking of the channels: relations and topic groups, encoded by connection and hue, were recognized almost universally, while size, the least accurate channel, was the most ambiguous; the typical alternative reading of size is "importance". The design responded with an explicit size legend and a sentence in the About overlay. The sample is small and not representative, so these results are indicative rather than conclusive.

5.4 Findings about retrieval

The visualization revealed properties of the underlying retrieval that are easy to state but hard to see in numbers:

- **Semantic ≠ spatial proximity:** Nearest neighbors in the 384-D space can be far apart in the projection (§4.2). Any 2D or 3D map of an embedding space is an approximation, and the query star makes this visible.
- **Retrieval follows meaning, not category:** 59 of 270 documents have at least three of their eight neighbors in another cluster. Document 112 (a graphics question about map projections) retrieves context from two clusters, which is semantically sensible.
- **Weak queries retrieve scattered context:** The documents with 8/8 cross-cluster neighbors are very short or generic ("Sigh.", "I have just a couple of questions about this technique."). Their embeddings carry little topical signal, so their neighbors are almost arbitrary, the same pathology that makes vague questions perform badly in production RAG systems.
- **Thresholds are corpus-specific:** Absolute cosine values of MiniLM on short forum posts are low (p99 = 0.41); the proposal's thresholds would have removed almost all edges.

5.5 Deviations from the proposal

Table 5. Deviations from the proposal and their reasons.

Proposal	Final implementation	Reason
Similarity thresholds 0.6 (default), tiers 0.6/0.7/0.85	Export 0.25, default 0.35, tiers 0.35/0.45/0.55	Empirical distribution (median 0.025, p99 0.41)
Edges only above threshold	+573 display-only knn links (schema v1.1)	A retrieval result must be shown completely
Structural edges as a separate set	Dual-typed edges (semantic + struct flag)	Most same-category pairs were already semantic edges
Default force-directed layout	Gentle forces seeded from UMAP	Default forces collapsed the cluster separation
Libraries from a CDN	Vendored, fixed version pair	Version conflicts in newer releases; offline demo
Slider range up to 1.0	Slider 0.25–0.75	Almost no edges above 0.7

One engineering lesson is worth recording. Recent releases of the 3D libraries made the CDN setup from the proposal unusable: the UMD build of three.js no longer defines a global, and the graph library's prebuilt renderer is incompatible with newer three.js materials. The solution was to vendor one tested version pair locally (three.js 0.169 and 3d-force-graph 1.73), which also made the application independent of the network on demo day.

6 Limitations and Future Work

- **Node selection as query:** The query is simulated by clicking an existing document. Free-text queries would require an embedding model in the browser (for example transformers.js) or a backend API; the architecture already supports this, because the query star needs only a list of neighbors.
- **Static, precomputed corpus** of 270 documents. Changing the corpus means re-running the pipeline and re-calibrating the thresholds.
- **Approximate projection:** Semantic neighbors can be spatially distant; the query star exposes but cannot remove this distortion.
- **Color-vision deficiency:** Red, orange and green clusters are hard to tell apart for deuteranopes and protanopes; position and labels compensate only partially.
- **Truncation:** The embedding sees only the first 256 word pieces of long documents.
- **Desktop only**, designed for about 300 nodes. Thousands of nodes would require instanced rendering and GPU-side similarity computation.

Natural extensions, deliberately left out of scope, are live free-text querying, semantic zooming with hierarchical cluster super-nodes, comparison of several embedding models in the same view, and chunk-level instead of document-level retrieval, which is how most production RAG systems work.

In summary, the project delivers what the proposal promised: a reproducible pipeline, a perceptually grounded 3D encoding and three interactions that make the retrieval step of a RAG system tangible in a single gesture. The most valuable outcomes were not planned: the visualization itself revealed where embedding retrieval behaves counter-intuitively.

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Appendix A — Reproduction

Requirements: Python 3.11 and a WebGL-capable desktop browser. From the repository root (Windows PowerShell):

```
py -3.11 -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r pipeline\requirements.txt
cd pipeline
python build_graph.py --recompute     # ~80 s, deterministic
python validate_json.py
cd ..\web
python -m http.server 8000           # open http://localhost:8000
```

The page must be served over HTTP, not opened as a file, because browsers block ES modules and data loading from local files. On hybrid-graphics laptops the browser should run on the dedicated GPU. Demo documents: id 112 (cross-cluster query), "Greetings, this is a general call for information…" (dense neighborhood) and "Forwarded from the Mars Observer Project" (peripheral sci.space document).